

HPC & shell quick reference (Ivy_Net)

Purpose: One page of copy-paste commands and “gotchas” we hit often in chat.

Depth lives elsewhere — follow links when you need full procedures.

Topic	Long-form doc
Git recipes (stash, rebase, recovery)	GIT_FOR_DUMMIES.md
Git mental model (Mac / HPC / GitHub)	GIT_MULTIPLE_MACHINES_ELEMENTARY.md
rsync, gitignore, what to pull/push	scripts/DATA_SYNC.md
Slurm logs, <code>tail -f</code> , <code>track_slurm.sh</code>	tenure/HPC_SLURM_PIPELINE_GUIDE.md
Faithful 537 sweep on Rivanna	sports/documents/RIVANNA_RUNBOOK.md
Faithful 537 step-by-step	sports/documents/Rivanna_Faithful_537_Sweep_For_Dummies.md
Faithful 538 sweep	sports/documents/Rivanna_Faithful_538_Sweep_For_Dummies.md
Doc index	README.md

Convention: Run Slurm and helper scripts from the **Ivy_Net repo root** (directory containing `sports/`, `tenure/`, `scripts/`, `sim_job.slurm`).

1. Two-second health checks

Git

```
cd ~/Ivy_Net
git status -sb
git branch -vv
```

Status line	Meaning
<code>[behind N]</code>	Run <code>git pull origin main</code> (or merge/rebase) before building on old code.
<code>[ahead N]</code>	Local commits not on GitHub yet — <code>git push</code> when ready.
<code>M / ??</code>	Uncommitted work — commit, stash, or discard before switching branches.

Rivanna queue

```
squeue -u $USER
```

Matplotlib (same Python Slurm uses for sweep merge/plots)

```
"$HOME/.conda/envs/sports_net/bin/python" -c "import matplotlib; print(matplotlib.__version__)"
```

If that fails, merge jobs that write `candidate_plots/` will fail or skip plots — install in `sports_net`, not only in an interactive notebook kernel.

2. Git — daily habit

```
cd ~/Ivy_Net
git pull origin main
# ... edit ...
git add <paths-you-mean>      # prefer explicit paths over blind `git add .`
git commit -m "Short description"
git push origin main
```

Normal `git push` does not overwrite GitHub `main`. Only `git push --force` (avoid on shared `main`) replaces remote history.

3. Git — local changes + need to sync

Uncommitted edits, origin moved

```
git stash push -u -m "WIP before sync"
git pull --rebase origin main
git stash pop
```

On a feature branch; bring in latest `main`

```
git fetch origin
git merge origin main
# fix conflicts if any, then git add ... && git merge --continue
```

Preview merge conflicts (optional)

After you have committed locally:

```
git fetch origin
git merge origin/main --no-commit --no-ff
# inspect: git status, git diff
git merge --abort # bail out without keeping the merge
```

4. Git — leave a branch, work on `main`

```
git switch main
git pull origin main
```

If the branch's work should stay in history, **merge or PR first**:

```
git switch main
git merge local-sweep-fixes
git push origin main
```

Delete the local branch

After merge into current branch:

```
git branch -d local-sweep-fixes
```

If Git refuses (`not fully merged` vs **remote** tracking branch) but `main` already has your commits:

```
git branch -D local-sweep-fixes
```

Optional remote cleanup:

```
git push origin --delete local-sweep-fixes
```

5. Git — accidental `chmod +x` on text files

Broad `chmod +x` on a folder can mark `.md` / `.txt` as executable. Git then shows **mode-only** diffs (`100644 → 100755`), not content changes.

```
chmod 644 scripts/DATA_SYNC.md scripts/requirements-notebook-mcp.txt
git status    # should be clean if only mode noise
```

At `git add -p`, type `n` for “Stage mode change?” on markdown unless you really want `+x`.

6. Slurm — submit jobs (not `bash job.slurm`)

Slurm scripts use `${SLURM_SUBMIT_DIR}`. Submit from repo root:

```
cd ~/Ivy_Net
sbatch sim_job.slurm           # Faithful 537 full pipeline
sbatch sim_job_538.slurm       # Faithful 538 full pipeline
sbatch pipe_job.slurm          # Tenure 540 papermill pipeline
```

Running `./sim_job.slurm` directly (without `sbatch`) often breaks under `set -u` because `SLURM_SUBMIT_DIR` is unset.

7. Slurm — monitor and cancel

```
squeue -u $USER
tail -f slurm_out/slurm-sim_job-<JOBID>.out
tail -f slurm_out/slurm-sim_job-<JOBID>.err
```

Helper (follow latest or a specific job's `.err`):

```
./scripts/track_slurm.sh
./scripts/track_slurm.sh <JOBID>
```

Cancel:

```
scancel <JOBID>
```

`.err` is not “errors only” — tqdm and many libraries log progress there even when the job succeeds.

8. Clean logs vs clean sweep outputs

Script	What it removes
<code>./scripts/clear_slurm.sh</code>	All <code>slurm_out/slurm-*</code> logs (and legacy root <code>slurm-*</code>)
<code>./scripts/clean_rivanna_faithful_537_sweep.sh</code> <code>--dry-run</code>	Lists 537 sweep tree + 537/sim_job logs only
<code>./scripts/clean_rivanna_faithful_537_sweep.sh</code> <code>--yes</code>	Deletes <code>rivanna_faithful_537/</code> + related logs
<code>... --yes --slurm-all</code>	Above + runs <code>clear_slurm.sh</code>

538 has an analogous clean script — `scripts/clean_rivanna_faithful_538_sweep.sh` (same `--dry-run` / `--yes` pattern).

Typical **fresh 537** run on Rivanna:

```
./scripts/clean_rivanna_faithful_537_sweep.sh --dry-run
./scripts/clean_rivanna_faithful_537_sweep.sh --yes
sbatch sim_job.slurm
```

9. Mac ↔ Rivanna sync (not Git)

Sweep **results** and Slurm logs are **gitignored**. Use rsync, not `git add`.

```
# Mac → Rivanna (sweep code only; not generated results)
./scripts/rsync_push_to_hpc.sh sweep

# Rivanna → Mac (results + logs)
./scripts/rsync_pull_from_hpc.sh sweep
./scripts/rsync_pull_recent_hpc.sh quick

# Preview
DRY_RUN=1 ./scripts/rsync_pull_from_hpc.sh sweep
```

After **code** changes on Mac: `git push`, then on Rivanna `git pull` in the same repo path **before** the next `sbatch`.
`rsync_push sweep` does **not** sync repo-root `sim_job.slurm` — use Git for that file.

Details: `scripts/DATA_SYNC.md`.

10. Faithful sweep outputs (537 / 538)

Artifact	Path
537 merged Stage 2	sports/outputs/simulation_sweeps/rivanna_faithful_537/stage2_results_merged.csv
537 grouped ranking	.../rivanna_faithful_537/grouped_candidates.csv
537 candidate plots	.../rivanna_faithful_537/candidate_plots/
538 (same layout)	.../rivanna_faithful_538/

Not sports/outputs/simulation_sweeps/candidate_plots/ at the sweep root — Rivanna merge writes under rivanna_faithful_537/ (or _538/).

Plots are created only in the final merge Slurm step (537_merge / 538_merge). Check that job’s log for:

- ==> candidate_plots: N PNG(s) -> ...
- or skipping candidate_plots/ - matplotlib is not installed

11. 537 notebook Cell 10 vs sweep CSV (knob alignment)

- Cell 10 widgets are authoritative for the playground plot; sim_config.py is defaults only — use Load defaults from sim_config.py after editing config.
- Promotions per run (K) and Pools (#) are widget sliders (not hidden in config only).
- Pool–talent bins (#) is aggregation bin count — do not confuse with promotions K.
- Faithful sweep PNG titles use the same labels as Cell 10 via sports/cell10_knob_catalog.py .

To reproduce a sweep row in the notebook: match widget labels to grouped_candidates.csv , set Binning → Pools: equal pool count, and align K, Pools (#), Runs, N, Pool–talent bins (#) with the CSV row.

Operator detail: sports/documents/537_Manual.md .

12. Common failure modes (from recent sessions)

Symptom	Likely cause	Fix
<code>SLURM_SUBMIT_DIR</code> / <code>set -u</code> error	Ran <code>.slurm</code> with bash instead of <code>sbatch</code>	<code>cd ~/Ivy_Net && sbatch sim_job.slurm</code>
No <code>candidate_plots/</code> after "successful" run	Merge never ran, or matplotlib missing in <code>sports_net</code>	Check <code>slurm-* *_merge*.err</code> ; install matplotlib in Slurm env
Empty <code>candidate_plots/</code> folder, 0 PNGs	<code>grouped</code> empty or type mismatch in plot loop	Check merge <code>.out</code> ; ensure consistent Stage 2 schema
<code>git pull</code> blocked	Uncommitted changes	Commit, stash, or discard
<code>git branch -d</code> refused	Branch not pushed / not merged to remote tracking ref	Use <code>-D</code> if <code>main</code> already has the commits
Git shows diff with 0 insertions	File mode <code>755</code> vs <code>644</code> only	<code>chmod 644 <file></code>
Old sweep rows after code change	Mixed shard CSVs from prior run	Run clean script, full re- <code>sbatch</code>
Merge <code>ValueError: Length of ascending ...</code> (538)	Bug in <code>grouped_candidates</code> sort columns	Fix sweep code; re-run merge after patch

13. Environment on Rivanna

```
module load miniforge
conda activate sports_net # or ENV_NAME=... on sbatch line
```

Slurm jobs use `$HOME/.conda/envs/${ENV_NAME:-sports_net}/bin/python`, not necessarily the kernel you use in Cursor Remote SSH.

Override for one submission:

```
ENV_NAME=sports_net N_STAGE1_SHARDS=64 N_SHARDS=64 sbatch sim_job.slurm
```

14. Where to add new snippets

When a command keeps coming up in chat:

1. Add a **one-line entry** to the table in §12 or a new § if it's a new category.
2. Put **full procedures** in the domain runbook (`RIVANNA_RUNBOOK.md` , `DATA_SYNC.md` , etc.) and **link** from here — avoid maintaining two long copies of the same checklist.